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Ref- ISA/IND/WT/120/2026 **Date: 0 4 February 2026**

Exoplanet Project
(For students with little or no Python)

❖ **Overall Philosophy**

"Understand the physics, extract at least one real parameter, and interpret it physically."
No one fails because they can't code — they fail only if they don't **measure anything**.

✓ **Allowed Tools (No Coding Required)**

Participants may use:-

1. Excel / Google Sheets
2. Desmos / GeoGebra
3. NASA Exoplanet Archive (web)
4. Lightkurve (web interface)
5. Plot digitizer (if needed)

1 – Choose & Understand

✓ **Task 1: Select ONE Planet**

From:-

1. TRAPPIST-1e
2. K2-18b
3. WASP-39b
4. Kepler-452b
5. Proxima b

Deliverable:-

- 1-page note:

Star type

1. Detection method
2. Orbital period
3. Why interesting?

2 – Theory Without Fear

✓ **Task 2: Core Physics (Must Do)**

They must compute by hand or Excel:

a) Planet Radius

$$R_p = R_* \sqrt{\delta}$$

b) Equilibrium Temperature

$$T_p = T_* \sqrt{\frac{R_*}{2a}}$$

c) Density

$$\rho = \frac{3M}{4\pi R^3}$$

✓ **Deliverable:**

1. One worked example with numbers
2. Earth comparison table

3 – Data Without Coding

✓ **Task 3A: Get Light Curve**

From:

1. NASA archive → download CSV
2. Or provided dataset by you

✓ **Task 3B: Plot in Excel**

They must:

1. Plot flux vs time
2. Identify transit dip
3. Estimate depth manually

✓ **Deliverable:**

1. Screenshot of plot
2. Estimated transit depth

4 – Interpretation & Present

✓ **They must answer:**

1. Is it rocky or gaseous?
2. Inside habitable zone?
3. Detectable atmosphere?

✓ **Deliverable:**

1. 15 page report on the style taught

❖ **MINIMUM PASS CRITERIA (Very Important)**

A participant **passes** only if:

1. They used **real data**
2. They computed **at least one parameter**
3. They included **at least two equations**
4. They gave **physical interpretation**

If any one missing → project is incomplete.

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✓ **MARKING RUBRIC (Simple & Fair)**

Component	Marks
Planet understanding	20
Physics calculations	30
Data plot	20
Interpretation	20
Presentation quality	10

WHAT MAKES THIS MODEL POWERFUL

✓ Even without Python, they still:

1. Use real astrophysical data
2. Apply real equations
3. Generate real plots
4. Think like scientists

✓ This is **far superior** to:

"Make PPT on exoplanets"

The One Instruction You Should Give Them

✓ **Write this on the project sheet:**

"This is not a descriptive project. You must calculate at least one real planetary property using real data."

That single line upgrades your project to **research-grade**.

"विकसित भारत @2047 – आत्मनिर्भर, सशक्त और उन्नत भारत"

भवदीय

निदेशक/Director
भारत अंतरिक्ष अकादमी/India space Academy
नई दिल्ली/New Delhi
निदेशक कार्यालय/**Director Office**
भारत अंतरिक्ष अकादमी/**India Space Academy**